

601.445/645

Practical Cryptographic Systems

Symmetric Cryptography II

Instructor: Matthew Green

Housekeeping

- Quiz!
- Remember, these are supposed to be easy
- If you did the reading and came to class
- If it wasn't, let us know on Piazza
- We will drop the lowest two scores

Chapter 5

Symmetric encryption and authentication

The most basic function of cryptography is to ensure confidentiality and authenticity of messages that are sent over a hostile communications channel. In the most common setting, this is done using *symmetric encryption*: this refers to the setting where both sender and receiver share the same key. Symmetric primitives tend to be preferred for most data protection, simply because they tend to be more efficient than alternatives like public-key cryptography. In this chapter, we'll discuss the three most common symmetric cryptographic functions: encryption (and *encipherment*), for protecting message confidentiality; message authentication (MACs) for protecting the authenticity of data; and *authenticated encryption* which combines the protections offered by both of those primitives.

In this chapter we'll confine ourselves to discussing these *keyed* primitives. However, any discussion of symmetric cryptography would be incomplete if we did not discuss hash functions. Since that's an intricate topic, we'll discuss those in Chapter 6.

5.1 High-level recommendations

Many cryptography books choose to dive into the weeds of symmetric encryption, starting with block and stream ciphers as well as a variety of legacy modes that you should never use. For completeness, I'll also cover to those details a bit later in this chapter. But sometimes you just need to encrypt something! The question you might have right off the bat is: *what tool should I use to do that?* I'm going to take this opportunity to answer that question first.

Housekeeping

- **Recordings and slides**
 - I'm trying to keep slides up to date but one or 1.5 weeks behind
 - Our recording equipment is malfunctioning
 - I genuinely don't know what to do about that

News

News

Hey there! You are using WhatsApp: Enumerating Three Billion Accounts for Security and Privacy

Gabriel K. Gegenhuber (University of Vienna, Faculty of Computer Science and UniVie Doctoral School Computer Science), Philipp E. Frenzel (SBA Research), Maximilian Günther (University of Vienna, Faculty of Computer Science), Johanna Ullrich (University of Vienna, Faculty of Computer Science), Aljosha Judmayer (University of Vienna, Faculty of Computer Science)

News

NEWS

MikroTik router flaws allow takeover without a password

by Pieter Arntz | September 8, 2026



CERT Polska warns that attackers are actively exploiting a chain of critical MikroTik RouterOS flaws to seize control of routers exposed to the internet.



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News

- **Microtik routers allow the user to authenticate with a public key**
- But the key needs to match one stored in the router
- However, the user can supply a copy of the key

NEWS

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News

- **Microtik routers allow the user to authenticate with a public key**
 - But the key needs to match one stored in the router
 - However, the user can supply a copy of the key
 - The router checks part of the key (modulus and type) but does not check the RSA exponent
 - An attacker can sign with exponent 1

NEWS

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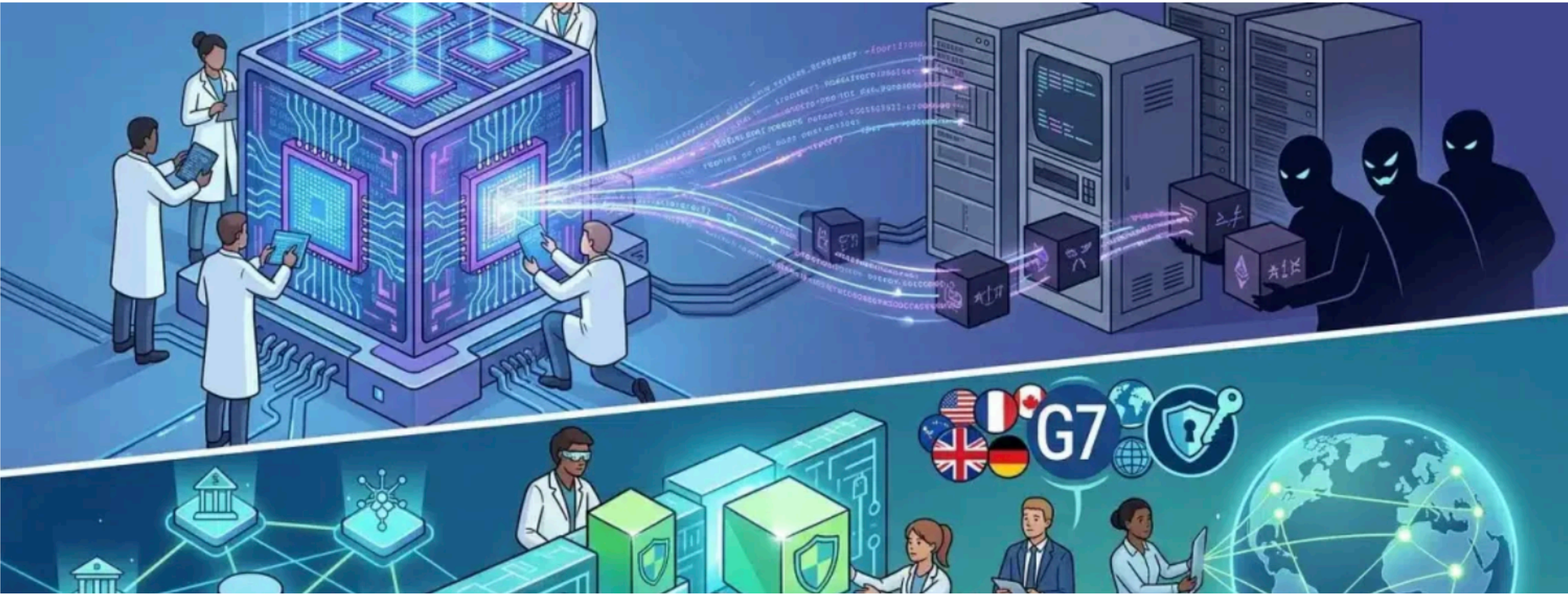
Home > Firewall Daily

G7, CISA Urge Urgent Shift to Post-Quantum Cryptography



by Samiksha Jain — September 7, 2026 in Firewall Daily, Cyber News

0



News Russian Hackers Are Silently Reading Your WhatsApp Messages Right Now

By Mauven MacLeod · 10 March 2026

♡ 21

Hello, Mauven here.

Yesterday, the Netherlands' two principal intelligence agencies issued a formal public warning. The AIVD, their domestic security service, and the MIVD, their military intelligence counterpart, confirmed that Russian state-sponsored hackers are running what they describe as a large-scale global campaign to compromise Signal and WhatsApp accounts.

Targets include government officials, military personnel, diplomats, and journalists. Dutch government employees have already been affected. Sensitive information has likely already been accessed.

I want to explain precisely why this matters to your small business.

The Attack Does Not Break the Encryption

This is the most important thing to understand first.

Signal and WhatsApp have robust [end-to-end encryption](#). That encryption has not been broken. The apps

Method Two: The Silent Linked Device

This one is considerably more insidious, because the victim never loses access to their account.

Both Signal and WhatsApp support a legitimate feature allowing users to link their account to multiple devices: a computer, a tablet, a second phone. On WhatsApp, this is called Linked Devices. The process normally requires scanning a QR code.

Attackers send targets a QR code or link disguised as a group invitation, a security verification prompt, or a resource from the app's official website. When the target scans it, they unknowingly link the attacker's device to their account.

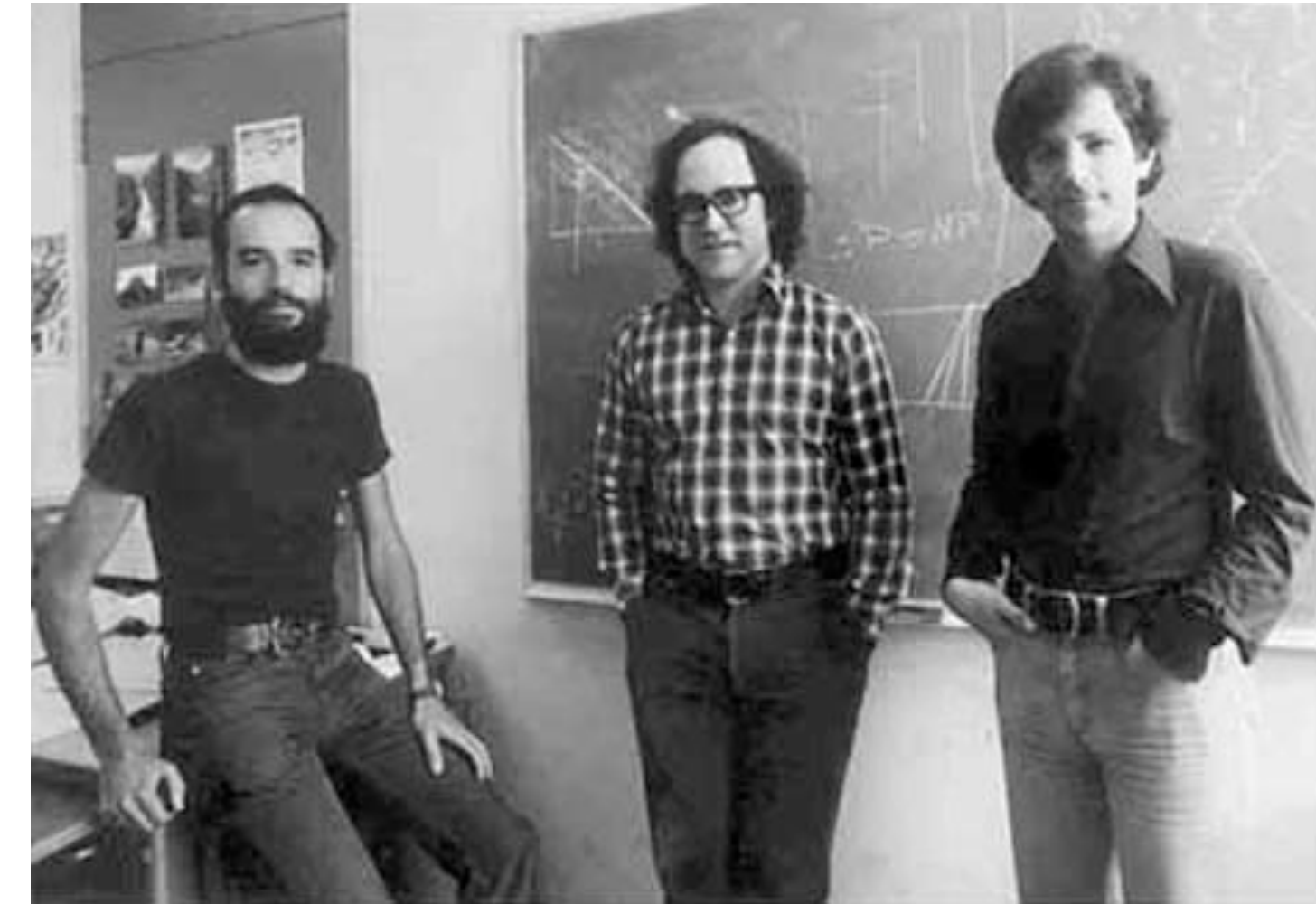
The target continues using WhatsApp normally. Their old messages are still there. Nothing appears to have changed. Meanwhile, the attacker's device is silently receiving every new message sent to or from that account, in real time, potentially indefinitely.

Germany's BSI advisory notes that Signal lists all linked devices under Settings, but that users rarely check this. That is a reasonable observation: most people have never opened that menu in their lives.

Review

- Last time:
 - We discussed one-time ciphers
 - Then we started to talk about block ciphers (DES)
 - What's a block cipher, what's inside DES

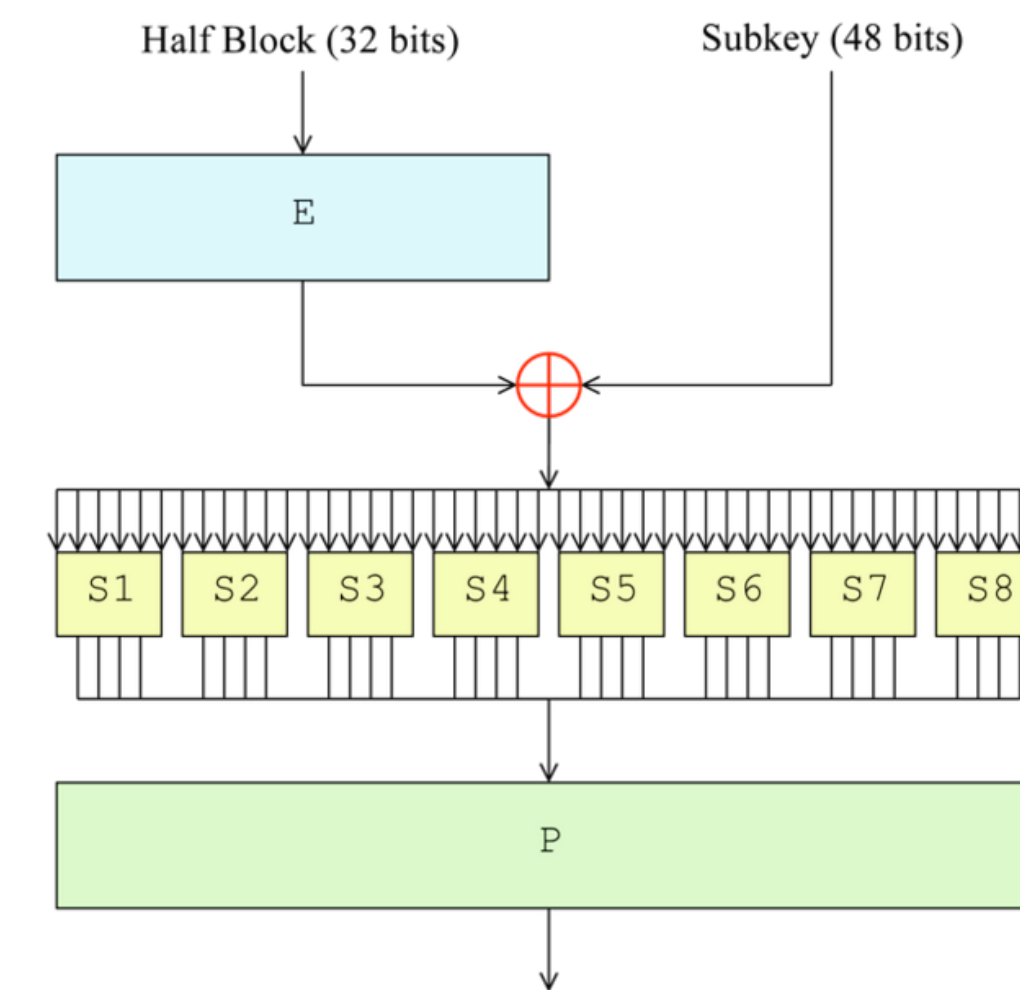
The 1970s



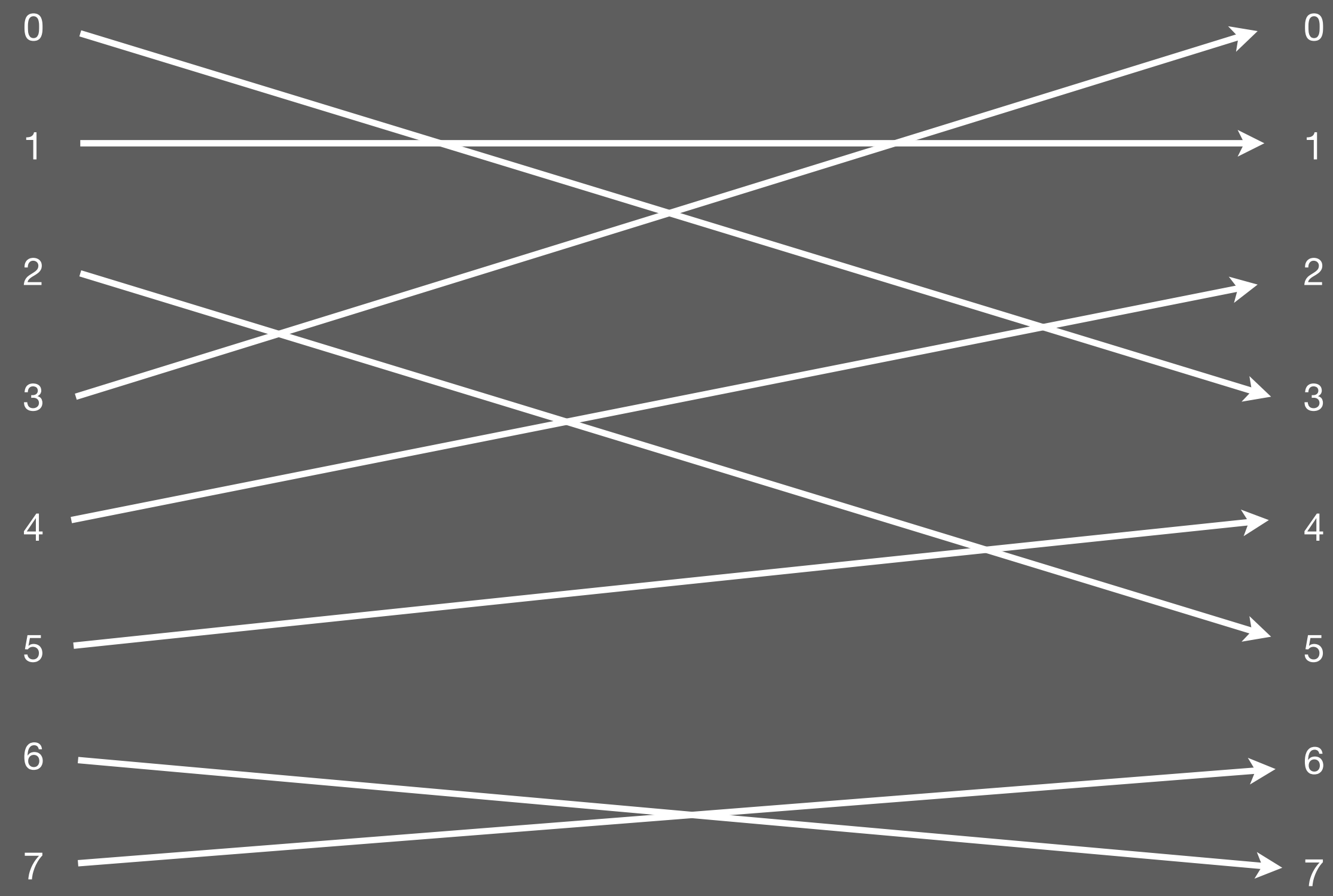
U.K. GCHQ

Data Encryption Standard

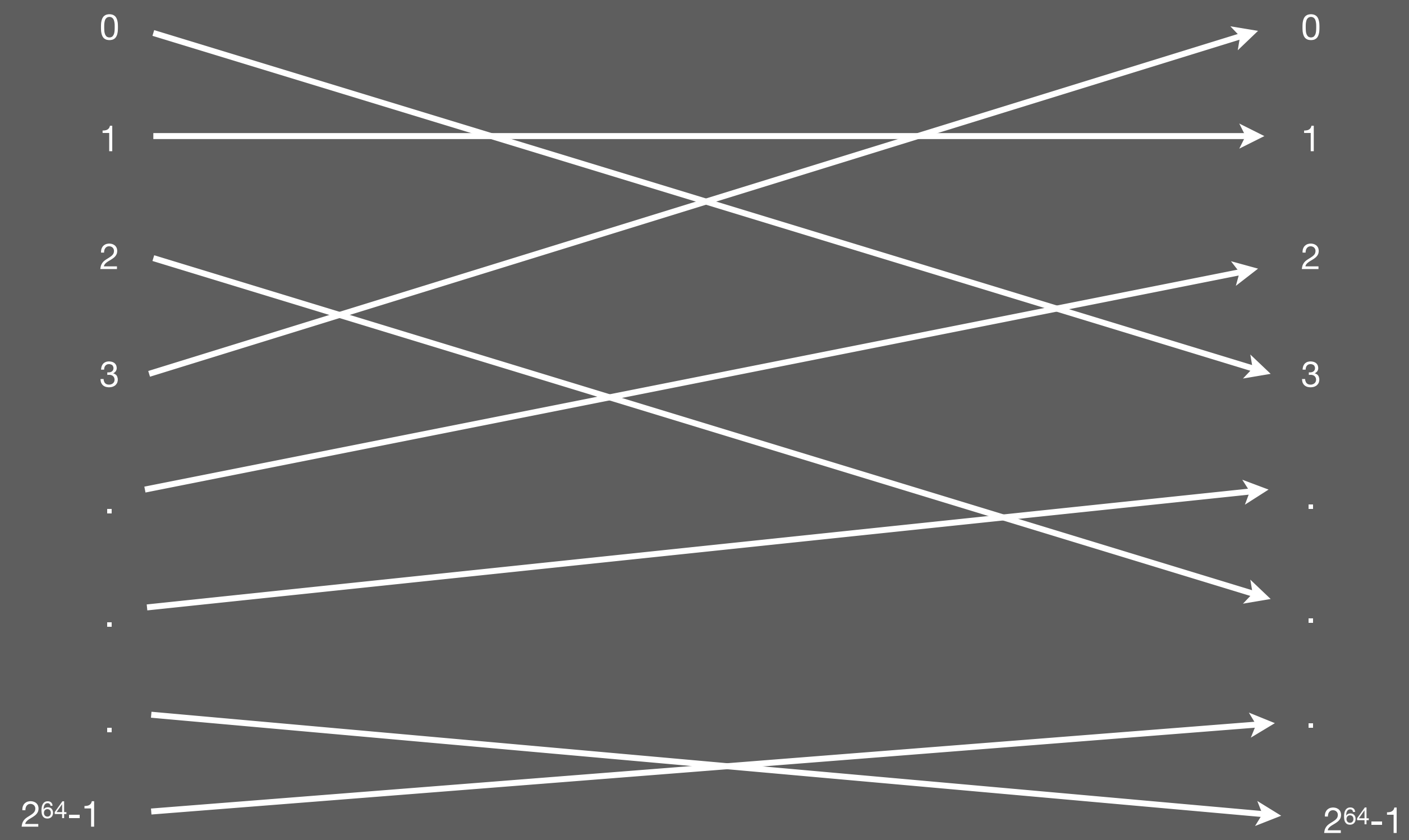
- Commercial-grade Block Cipher
 - 64-bit block size
 - 56 bit key (+ 8 bits parity)
 - “Feistel Network” Construction



Permutation



Permutation



Permutation Families

- Can't have just one permutation
 - Alice & Bob know the permutation
Adversary doesn't
 - Permutation is “random” (ish)
 - But there are 2^{64} possible permutations
 - DES has a 56 bit key...

Block Cipher

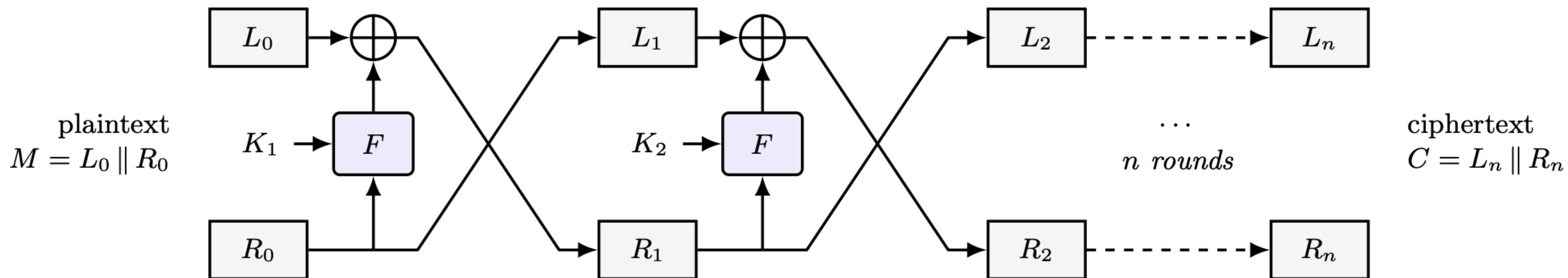
- Block cipher is a family of permutations
 - Indexed by a key (DES = 56 bit key)
 - “Pseudo-random”

Block Cipher

- Block cipher is a family of permutations
 - Indexed by a key (DES = 56 bit key)
 - Ideally: “Pseudo-random permutation (PRP)”

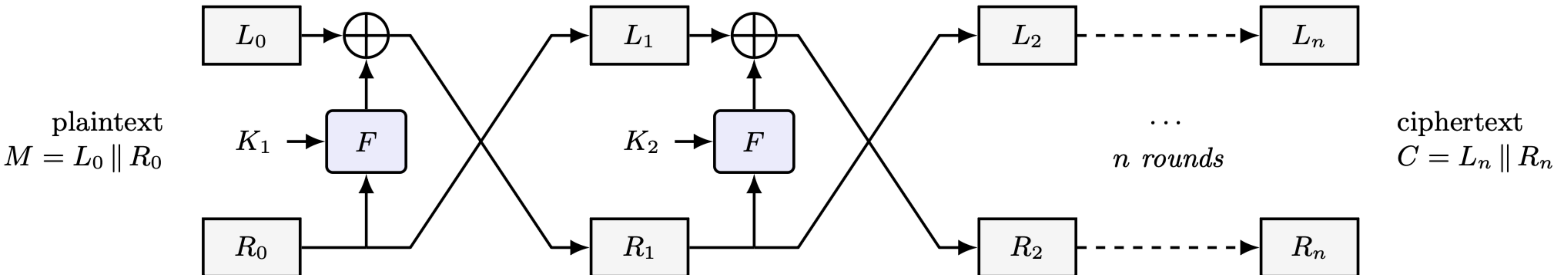
(i.e., attacker who does not know the key can't determine whether you're using a random permutation, or a PRP)

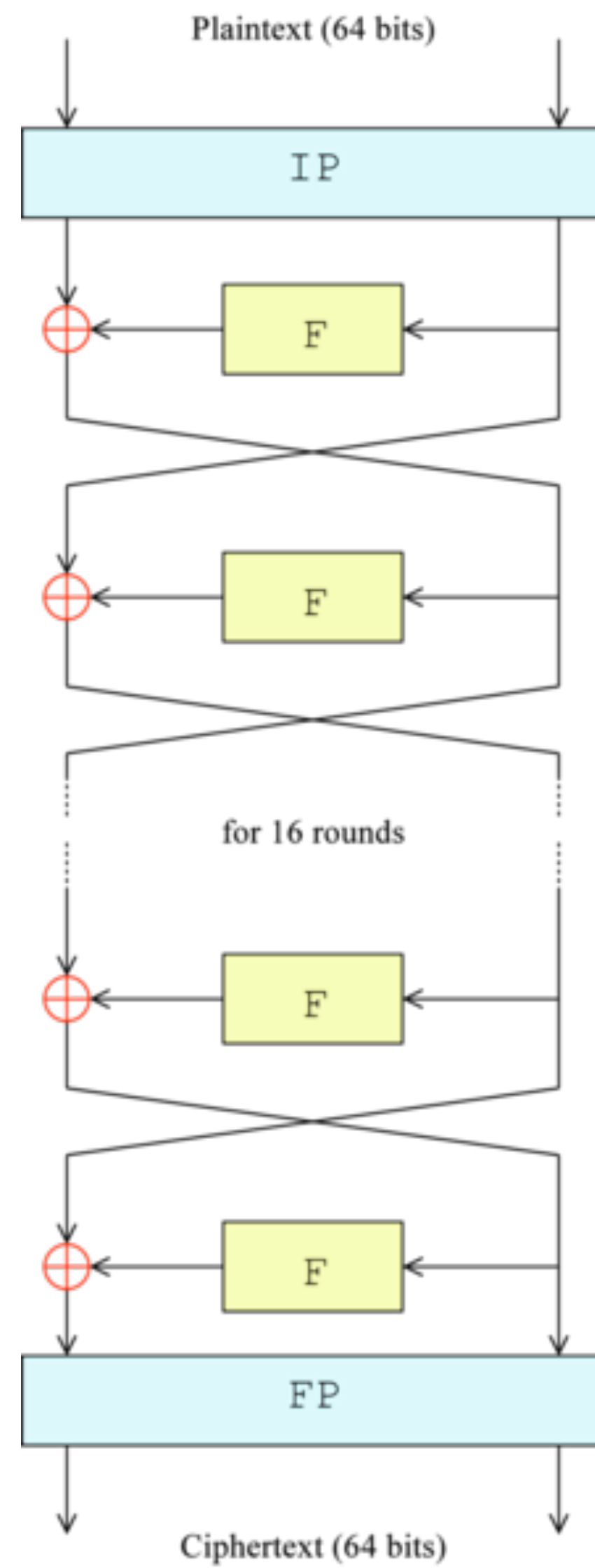
Block Cipher



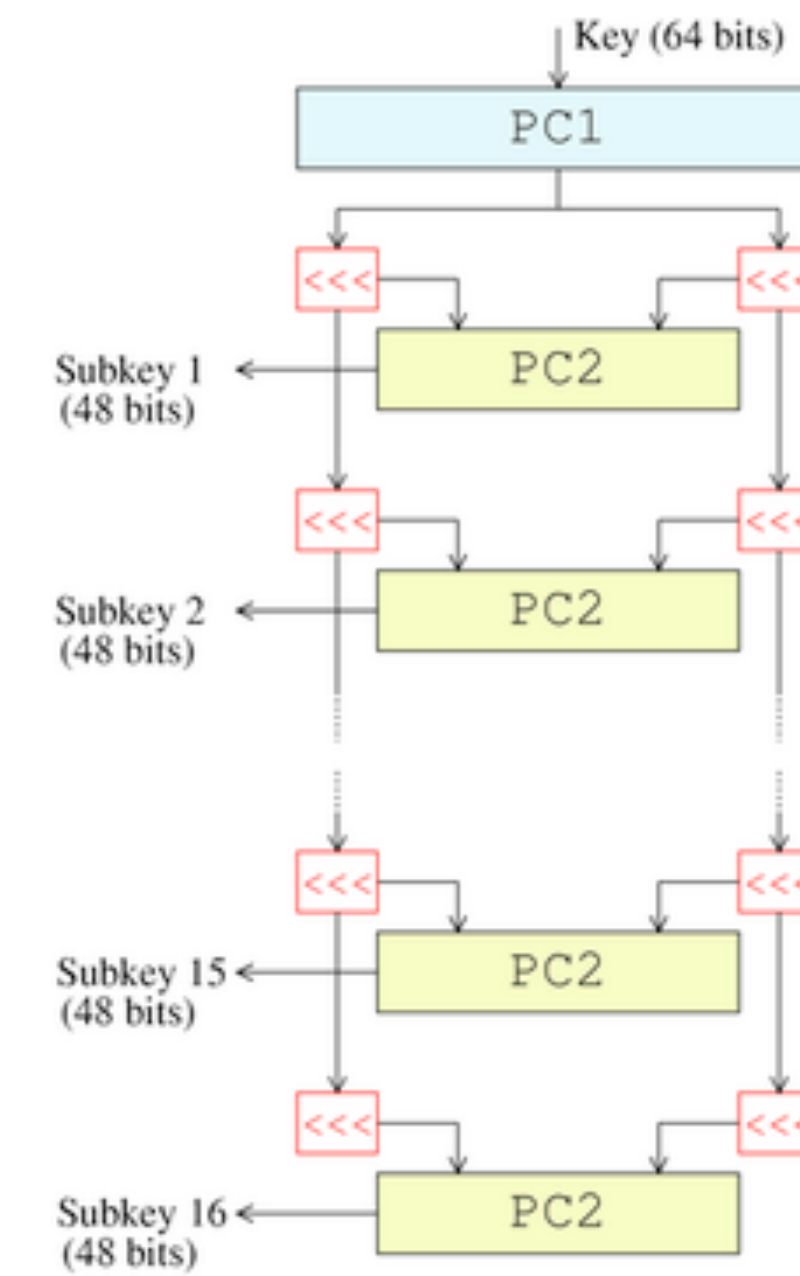
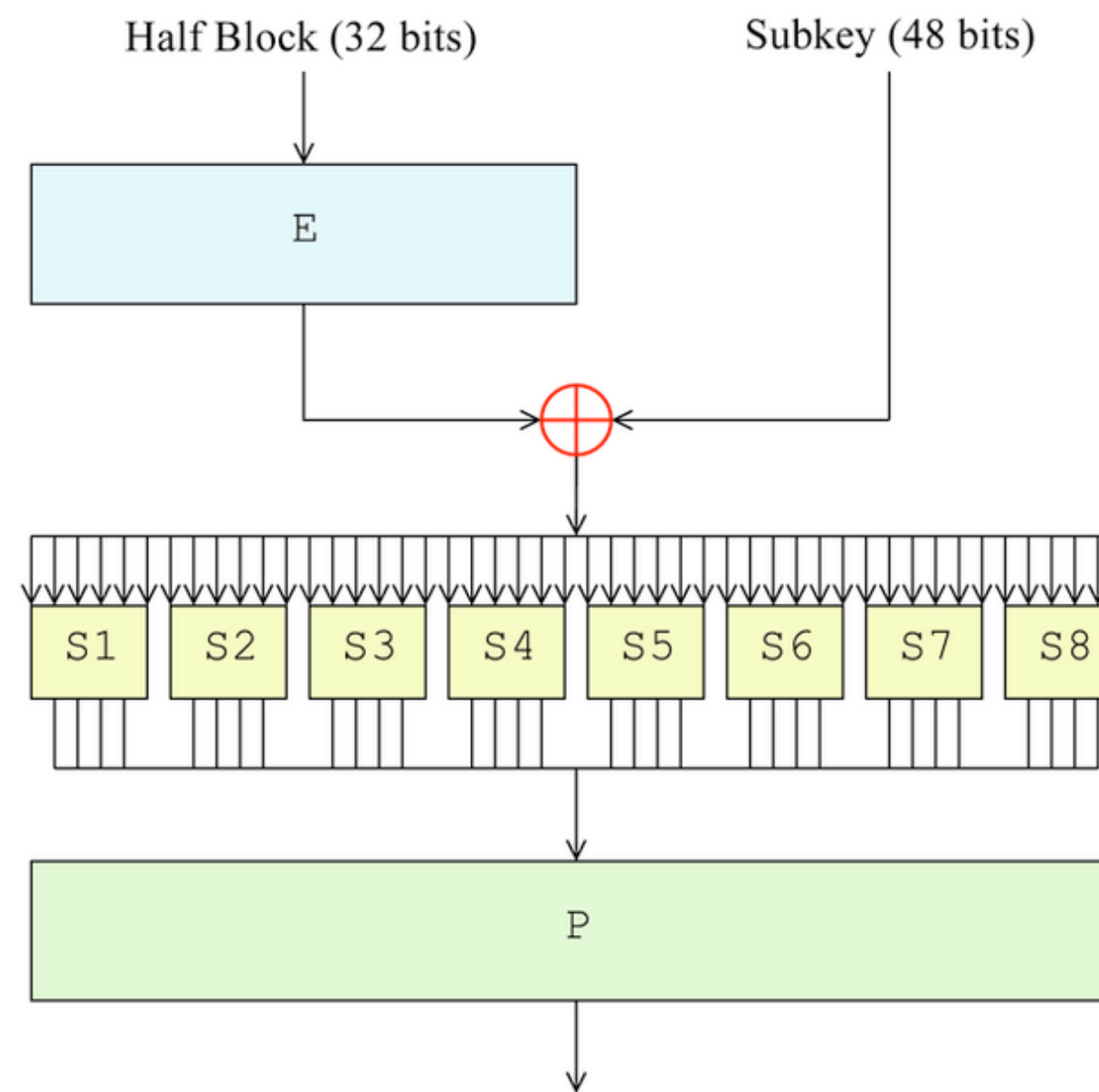
Block Cipher

- Neat result about Feistel networks (Luby & Rackoff)
 - If the function F is a pseudorandom function
 - And each instance of F has its own unique key
- Then the Feistel structure is a pseudorandom permutation



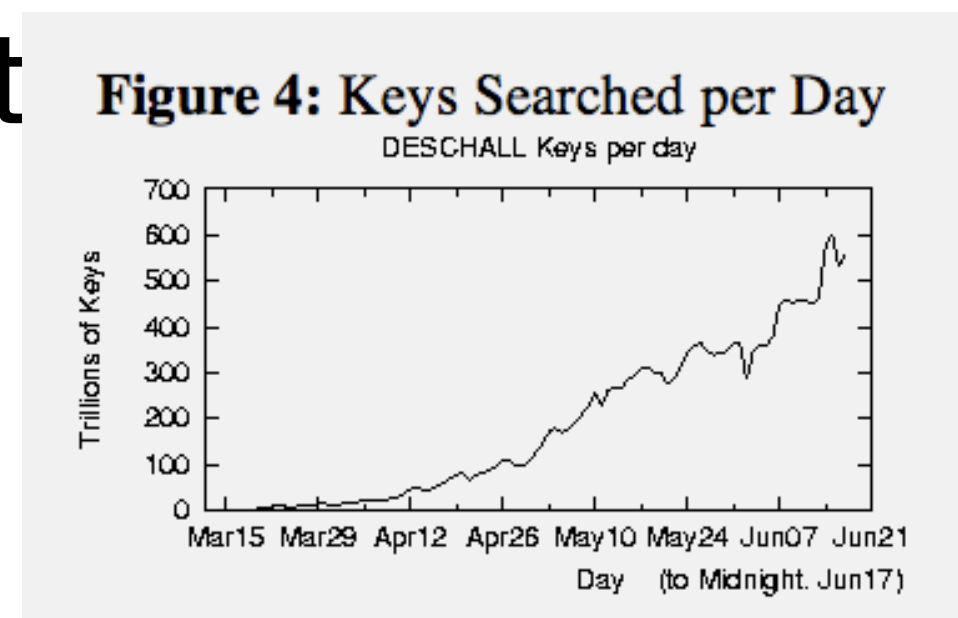


DES: 64-bit Block, 56-bit Key



DES

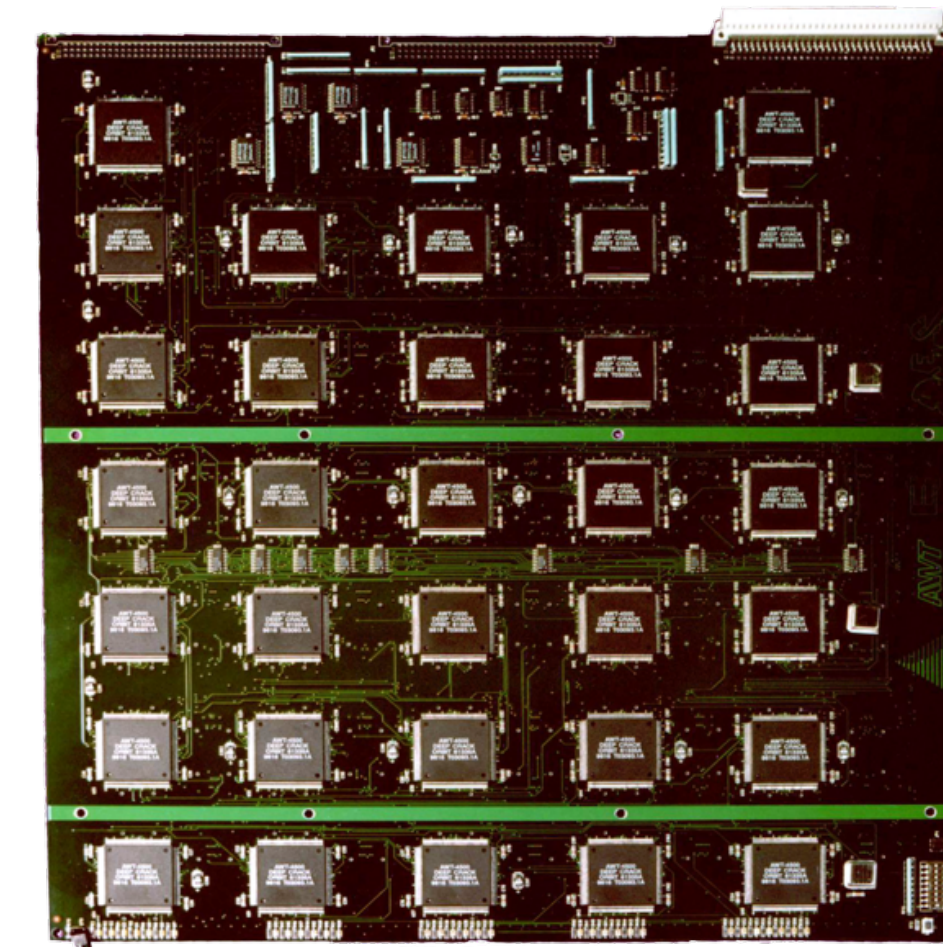
- Some “clever” attacks on DES
 - However: practical weakness = 56 bit key size
 - Practical weakness = 56 bit key size
 - Practical weakness = 56 bit key size
- DES (now being deprecated)



U.S. Data-Scrambling Code Cracked With Homemade Equipment

By JOHN MARKOFF

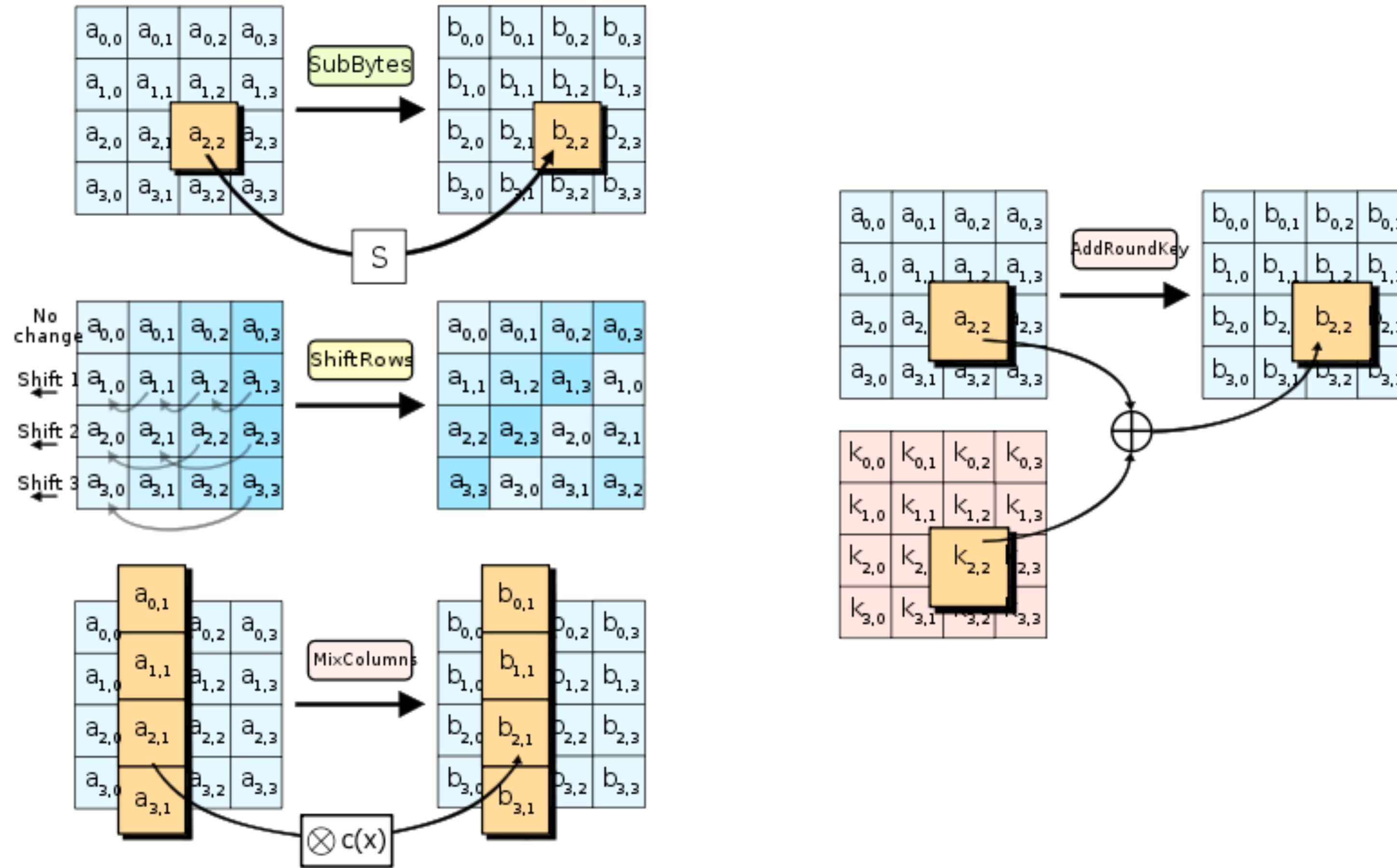
SAN FRANCISCO -- In a 1990s variant of a John Henry-style competition between man and machine, researchers using a homemade supercomputer have cracked the government's standard data-scrambling code in record time -- and have done it by out-calculating a team that had harnessed thousands of computers, including some of the world's most powerful.



AES

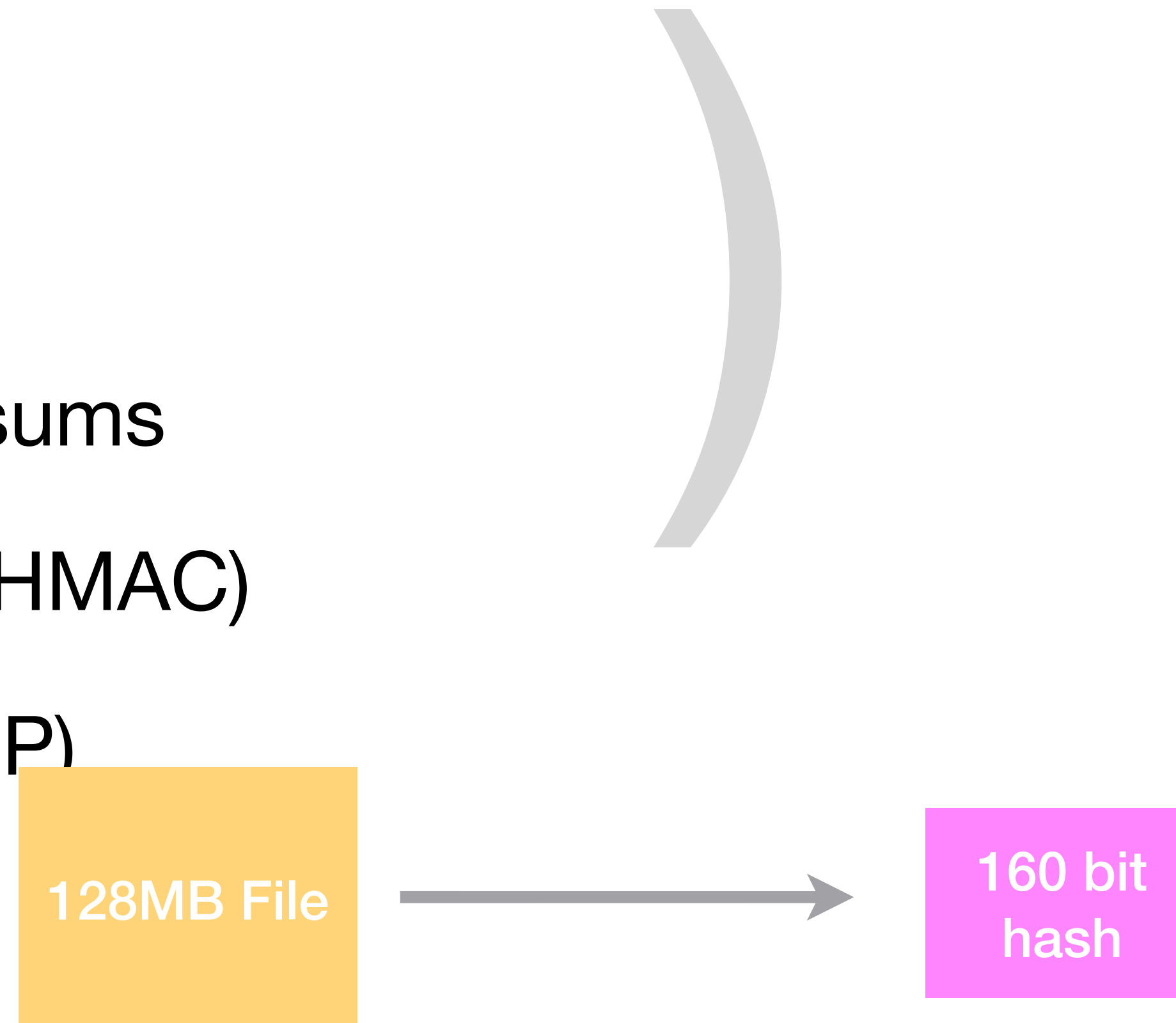
- NIST open competition:
 - Fast in software & hardware
 - Larger block size (128 bit)
 - Longer keys (128/192/256-bit)
- 5 finalists:
 - MARS, RC6, Rijndael, Serpent, and Twofish

AES: 128-bit Block, 128/192/256-bit Key



Hash Functions

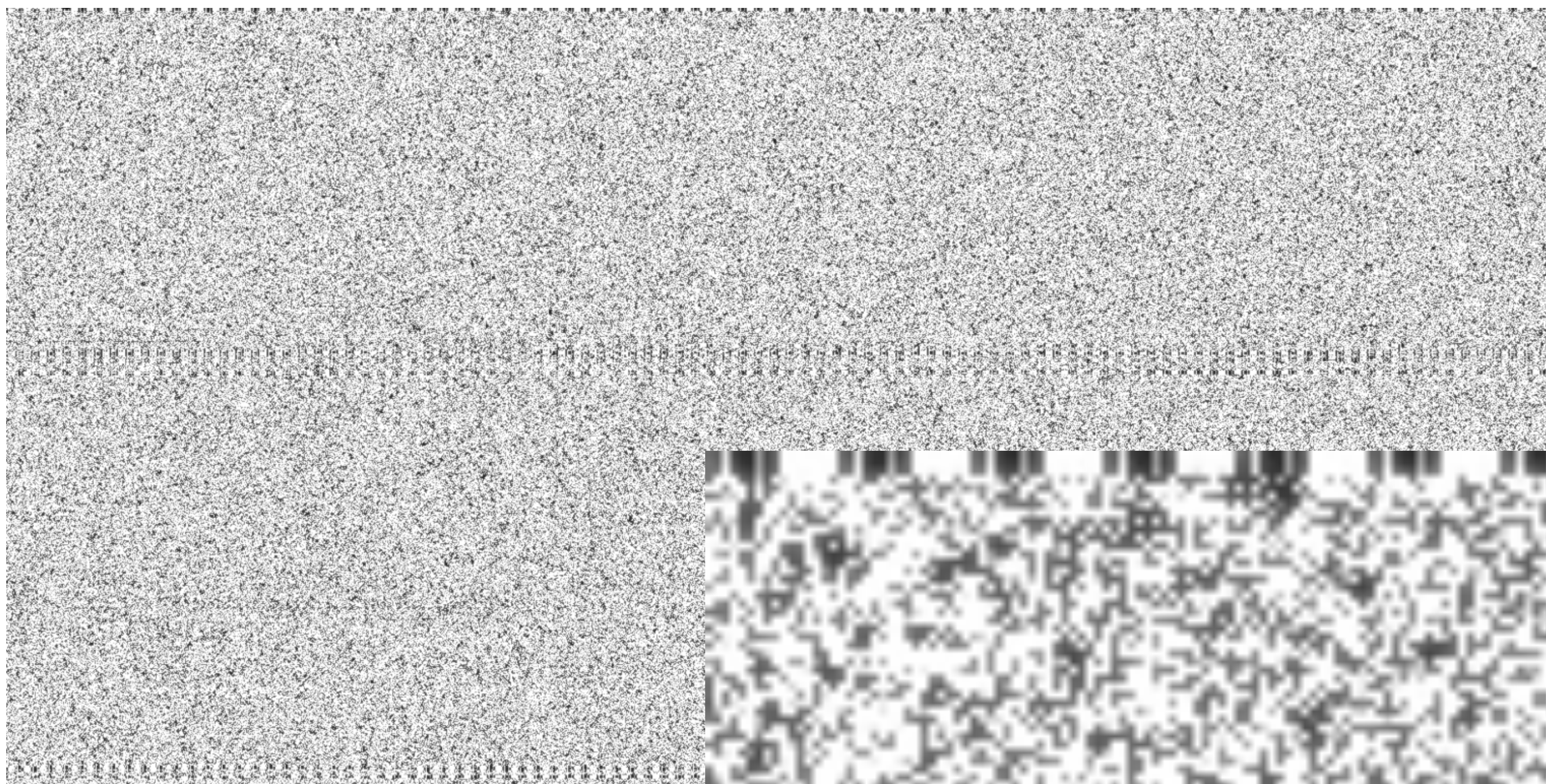
- Convert variable-length string to small “tag”
 - Hash tables
 - Signatures
 - Software checksums
 - MAC functions (HMAC)
 - Encryption (OAEP)



Review

- ECB Mode: Encrypt each block separately
- Problems?





ECB Mode

- ECB is deterministic
- Leads to problems, e.g.,:



Security of Encryption

- Semantic Security
 - Due to Goldwasser & Micali (1980s)
 - Informally: An encryption scheme is secure if adversary who sees ciphertext “learns as much” as adversary who doesn’t see ciphertext.
- Even if adversary can request chosen plaintexts
 - How do we state this formally?

Review

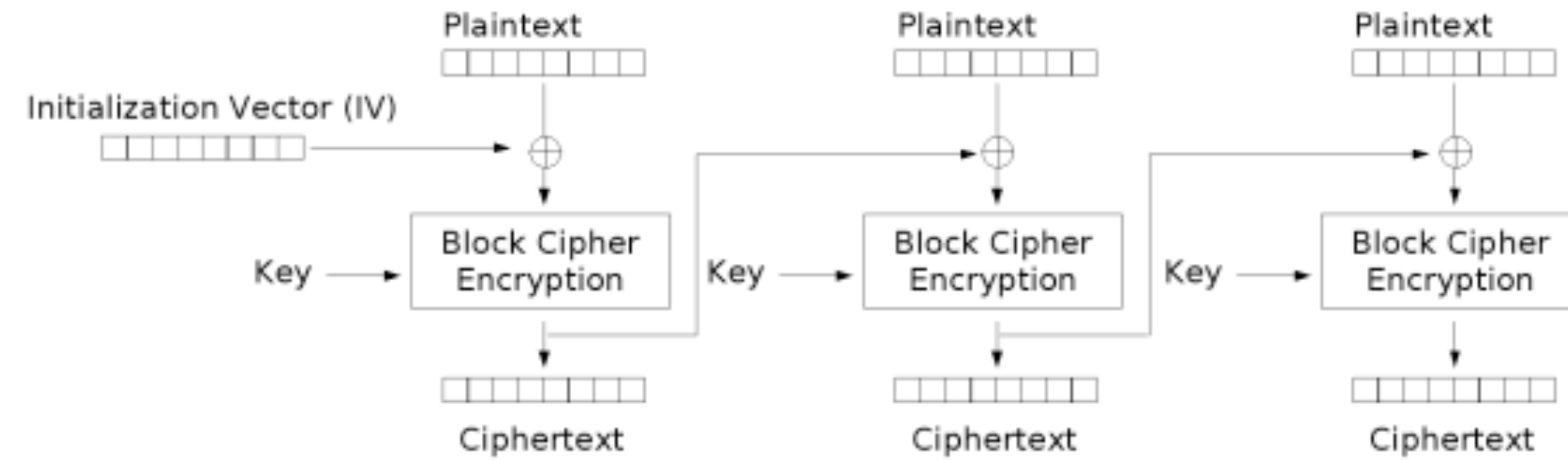
- Semantic Security (IND-CPA)



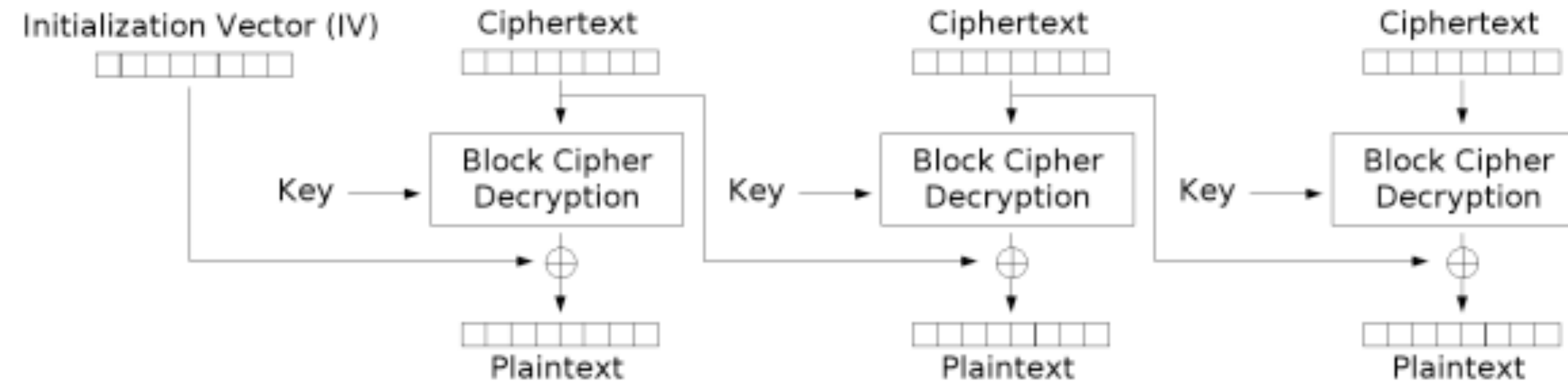
Using Block Ciphers

- ECB is not semantically secure, hence we use a “mode of operation”
 - e.g., CBC, CTR, CFB, OFB (and others)
- These provide:
 - Security for multi-block messages
 - Randomization (through an Initialization Vector)

CBC Mode



Cipher Block Chaining (CBC) mode encryption



Cipher Block Chaining (CBC) mode decryption

Security of CBC

- Is CBC a secure encryption scheme?
 - Yes, assuming a secure block cipher
 - Correct (random) IV generation
 - Can prove this under assumption that block cipher = Pseudo-Random Permutation (PRP)
- Bellare, Desai, Jorikpää & Rogaway (2000)
 - Easy to use wrong...
 - Most important: use a unique & random IV!

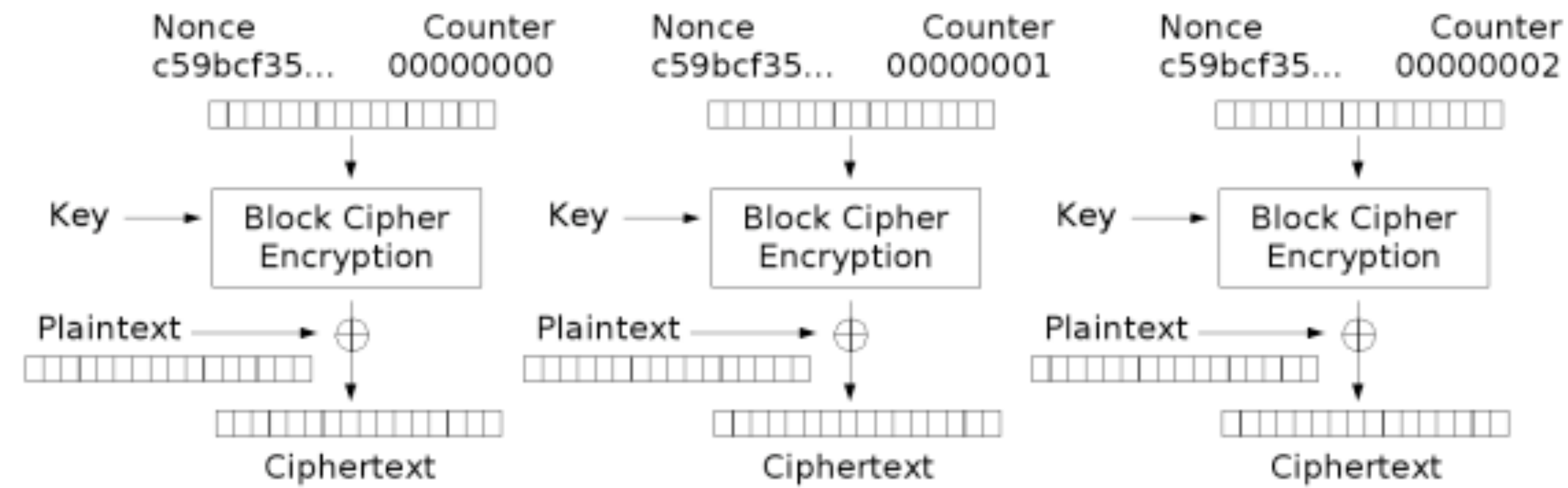
The size of the frame of data to be encrypted or decrypted (i.e. how often a new CBC chain is started) depends on the particular application, and is defined for each in the corresponding format specific books of this specification. Unless otherwise specified, the Initialization Vector used at the beginning of a CBC encryption or decryption chain is a constant, iv_0 , which is:

0BA0F8DDFEA61FB3D8DF9F566A050F78₁₆

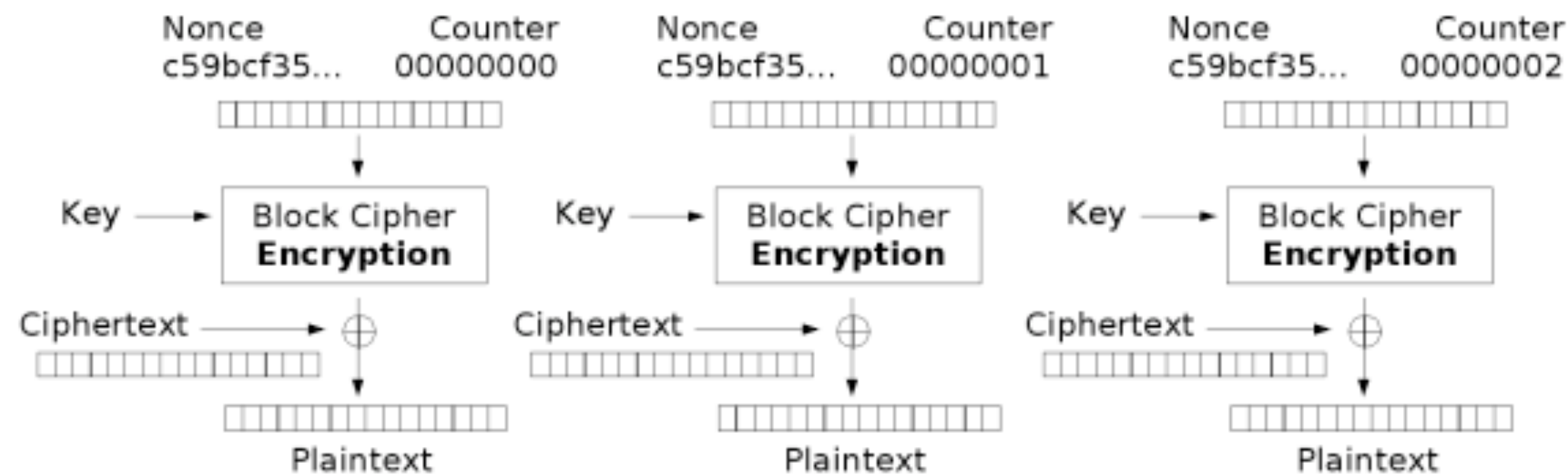
Advanced Access Content System (AACCS)

Introduction and Common Cryptographic Elements

CTR Mode



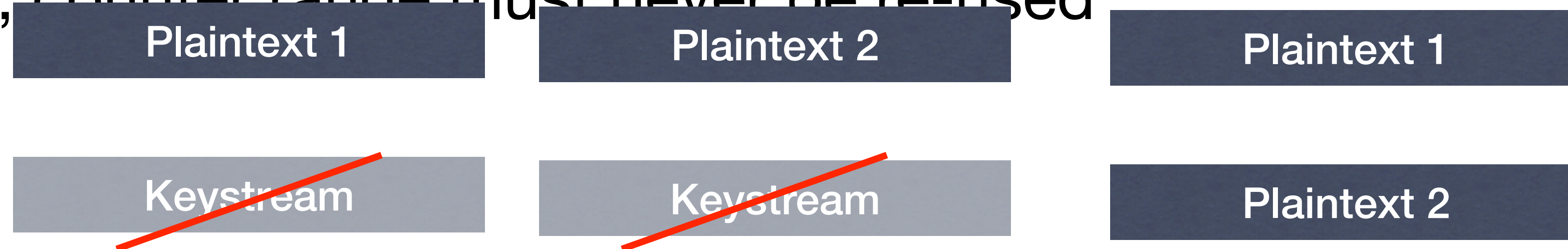
Counter (CTR) mode encryption



Counter (CTR) mode decryption

Security of CTR

- Yes, assuming secure block cipher (PRP)
- However, counter range must never be re-used



- Similar example: MS Word 2003
 - (they used RC4, but same problem)

Point of order

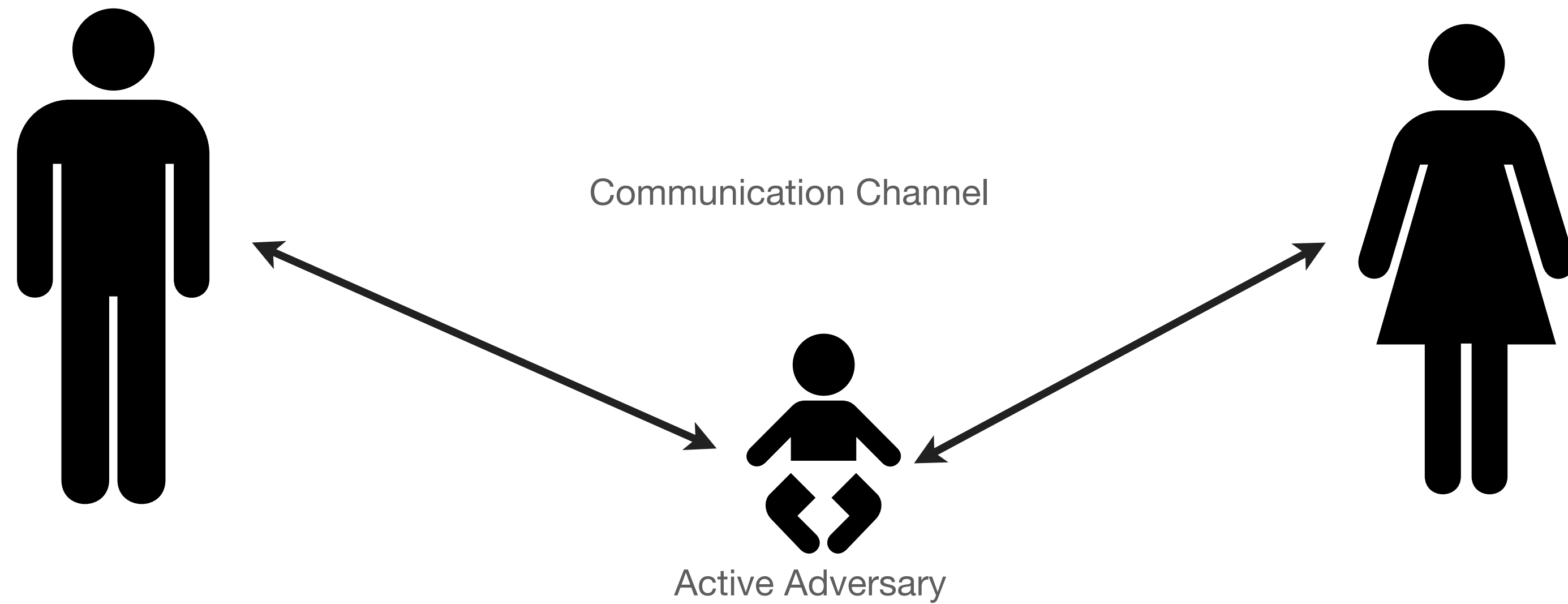
- Proofs of security:
 - We don't know how to prove that DES or AES are secure block ciphers
 - But if we assume that the block ciphers are secure PRPs then:
- We can prove that CBC & CTR & OFB & CFB etc. are secure encryption modes.

<http://www.cs.ucdavis.edu/~rogaway/papers/sym-enc-abstract.html>

Malleability

- The ability to modify a ciphertext
 - Such that the plaintext is meaningfully altered
 - CTR Mode (bad)
 - CBC Mode (somewhat bad)

Authenticated Encryption



MACs

- Symmetric-key primitive
 - Given a key and a message, compute a “tag”
 - Tag can be verified using the same key
 - Any changes to the message detectable
- To prevent malleability:
 - Encrypt then MAC
 - Under separate keys

MACs

- Definitions of Security
 - Existential Unforgeability under Chosen Message Attack (EU-CMA)
- Examples:
 - HMAC (based on hash functions)
 - CMAC/CBC-MAC (block ciphers)

Authenticated Encryption

- Two ways to get there:
 - Generic composition
Encrypt (e.g., CBC mode) then MAC
- two different keys, multiple primitives
 - Authenticated mode of operation
- Integrates both encryption & authentication
- Single key, typically uses only one primitive (e.g., block cipher)
- Ex: CCM, OCB, GCM modes

